An alternating group on the pairs

Human Compression project: a component of the L044 recovery

Let $G = A_{2m} = \langle \rho_0, \dots, \rho_{r-1} \rangle$ in its natural action, with a string C-marking. Suppose a maximal omission $H = G_i$ is transitive and preserves m pairs, and its induced block group is A_m. Throughout $m \geq 8$. We assume explicitly the rank bound for natural alternating groups of every smaller degree $12 \leq q < 2m$. Degrees below twelve use the acquired complete finite maxima. These are the strong-induction premises, not the theorem at degree $2m$.

We acquire the complete L047 block decomposition, its exact transitive-rank theorem and finite exceptions, the extremal independent-string theorem for symmetric groups, and the accepted lemma on centralizers of primitive actions on pairs. In particular, in a putative case $r \geq m$ the minimal block-generating family L is a connected interval, its commuting remainder C has size at most one, and its other remainder R has size at most one. The precise actual-action proof of this last fact is retained.

Write $V = \mathbb{F}_2^m$, $z = (1, \dots, 1)$, and let E be the even-weight subspace. The acquired elementary module argument says that the A_m-invariant subspaces of V are $0, \langle z \rangle, E, V$ for $m \geq 5$: a three-cycle applied to a nonconstant vector gives a difference of weight two, and conjugating such a difference generates E. The kernel of an even permutation group on the pairs lies in E.

Lemma 1. *Let $m \geq 8$ be even. Let $Q \leq A_m$ be intransitive and have a string C-marking of rank $m - 3$. Suppose this marking arises faithfully from a permutation group T on $2m$ points preserving m pairs, with block image Q. Assume the stated smaller-degree alternating rank bounds. Then every even involution centralizing T preserves the pairs.*

Proof. For an intransitive group on orbit sizes $a_1, \dots, a_q$, successive projection and kernel normalization give independent length at most $\sum(a_j - 1) = m - q$, by Whiston's bound. Equality forces the full product of the symmetric groups on those orbits. Indeed, choose a minimal subfamily generating one projection and normalize the remaining members into its kernel by words in that subfamily. Equality forces both bounds to be attained; Whiston's equality clause identifies the first projection and the last kernel as the full symmetric factors. Iterate over the remaining orbits. This also handles singleton orbits, whose contribution is zero.

Consequently Q has at most three orbits. Three would force that full product, containing an odd permutation, contrary to $Q \leq A_m$. Thus there are exactly two orbits, of sizes $a \leq b$, with $a + b = m$. If $a = 1$, the faithful action on the other $b = m - 1$ points has rank $b - 2$. Here $b \geq 7$ is odd. The complete acquired L047 theorem excludes every proper transitive group other than A_b, S_b at this rank; its proper high-rank exceptions have even degree. The even condition excludes S_b, and the assumed or finite alternating bound excludes A_b. Hence $a \geq 2$.

Projecting onto the a points and normalizing the other generators into the kernel gives

$$m - 3 \leq \mu(Q|_a) + \mu'(A_b) \leq (a - 1) + (b - 2) = m - 3.$$

Here μ is maximum irredundant generating length and μ' maximum independent length. Equality makes the first projection S_a. Reversing the two projections makes the other S_b. Thus Q is subdirect in $S_a \times S_b$. If $b \geq 5$, its kernel in the second factor is a normal subgroup of S_b contained in A_b, hence is 1 or A_b. The first possibility would identify Q with S_a, of independent

length at most $a-1<m-3$. Therefore that kernel is A_b. Projection onto the first factor and the parity condition now identify the full group:

$$Q=(S_a\times S_b)\cap A_m. \tag{1}$$

Consider first $a=3$, so $b=m-3$ is odd and $b\geq 5$. The kernel of the projection of (1) onto S_b is $A_3=C_3$. Choose a minimal distinguished family generating S_b. The normalization inequality and rank b force its size $b-1$. Its images are an extremal independent involutory string. The acquired theorem makes them a natural Coxeter path, for $b\geq 7$; its complete degree-five boundary gives the same conclusion for $b=5$. They occupy an interval of the string. The one remaining mark commutes with the whole path except possibly an endpoint, so its S_b image centralizes the natural S_{b-1} fixing an endpoint. That centralizer in S_b is trivial. The remaining involution would therefore lie in C_3, impossible. This excludes $a=3$.

For $b\geq 5$ and $a\neq 3$, the normal subgroup $1\times A_b$ of Q, regarded as a subgroup of the faithful lift T, fixes all $2a$ points over the first orbit. Its action there maps the perfect group A_b to the abelian group of pair flips and is therefore trivial. It has no fixed point on the other $2b$ points, since it is transitive on their pairs. Its fixed set is exactly the first $2a$ points. A permutation centralizing T consequently preserves both of these sets. On the $2b$ points the block image is S_b, so the acquired pair-centralizer lemma says that every centralizer preserves the pairs. Its exceptional regular prime-dihedral group cannot be S_b for $b\geq 5$. Such a centralizer induces a permutation centralizing S_b on the blocks, hence induces the identity. Its remaining flip vector is constant. Thus its restriction there is either 1 or z_b, the flip of all b pairs.

If $a\geq 4$, the same pair-centralizer lemma applies on the $2a$ points, with block image S_a, and finishes the proof. If $a=2$, the kernel A_b fixes these four points, so their induced group is the faithful S_2 interchanging the two pairs. Its involution is one of the two double transpositions which interchange those pairs. The centralizer of such a double transposition in S_4 is dihedral of order eight; its even subgroup is the Klein four group of double transpositions, which preserves every partition into pairs. Since $b=m-2$ is even, both possible centralizers $1, z_b$ on the other points are even. An even involution centralizing T must therefore be even on these four points as well, and preserves their pairs.

The only remaining case has $b<5$, so $m=8$ and $a=b=4$. The two projections of Q are S_4. Their kernels are normal subgroups contained in A_4. Neither is trivial, as otherwise $Q\cong S_4$ would have five independent marks. Thus each is V_4 or A_4. We claim that the first kernel fixes pointwise the eight points over the other block orbit. Its action there is a homomorphism to $\mathbb{F}_2^4$, equivariant under Q. For A_4 it is trivial, since $A_4^{\mathrm{ab}}=C_3$. For V_4, conjugation by the first projection S_4 acts irreducibly on its two-dimensional binary space, so a nonzero image would have dimension two. Such an image would be invariant under the second projection S_4 acting by coordinate permutation on $\mathbb{F}_2^4$. No such subspace exists: an invariant subspace containing a vector of weight two contains the entire three-dimensional even subspace; a vector of weight one or three generates the full space; the only remaining nonzero vector is the constant vector. This proves the claim. The first kernel has no fixed points over its own four-block orbit, because it is transitive on those four blocks. Its fixed set is therefore exactly the other eight points. Centralizers of T preserve both sets. The pair-centralizer lemma, with block image S_4 on each set, now proves preservation of all eight pairs. □

Lemma 2. *Let $Q=(S_4\wr C_2)\cap A_8$ in its natural eight-point action. Every permutation centralizing a faithful lift T of Q to a permutation group on eight pairs preserves those pairs.*

Proof. The centralizer of Q on its eight letters is trivial. Indeed its subgroup $A_4\times A_4$ distinguishes the two four-point sets by its two normal factors; on either set, the centralizer of the natural A_4 is trivial. If T has two point orbits, each meets every pair once. The two row actions are the literal Q action. Its trivial centralizer makes the unique equivariant row interchange the one matching the original pairs, as required.

If T is transitive, identify it with Q. A point stabilizer J has index two in the stabilizer M of an original pair. In the eight-letter action,

$$M = (S_3 \times S_4)^+,$$

where the superscript denotes equal signs in the two factors. Its derived subgroup is $A_3 \times A_4$. To see equality, commutation of the first A_3 by a lift of a transposition gives all that A_3; the derived subgroup projects onto A_4 in the second factor, since the second projection is S_4. Removing the first coordinate therefore gives all of the second A_4. The opposite containment follows from the signs. Thus $M^{\mathrm{ab}} = C_2$ and its unique index-two subgroup is $J = A_3 \times A_4$. This subgroup has orbits of sizes $1, 3, 4$ on the eight letters. Its normalizer in Q fixes the unique singleton, so $N_Q(J) = M$. The coset description of a transitive centralizer then places every equivariant image of the chosen point in the same original pair. Equivariance preserves all pairs, proving the claim. □

Theorem 3. *Under the standing hypotheses and explicit smaller-degree induction, $r \le m - 1$.*

Proof. Assume $r \ge m$, and write $B = \langle L \rangle$. The images of L are independent in A_m, so $|L| \le m - 2$. Put $N = B \cap V$ and $K = H \cap V$. If C is nonempty, its unique mark is z; then m is even. IP gives $z \notin B$, so the module list makes $N = 0$. The faithful projection would give a rank-$|L|$ C-marking of A_m. Moreover, if R were nonempty, $\langle L, R \rangle$ would strictly contain B, with the same block image, and have a nonzero invariant kernel. For even m every such subspace contains z, contradicting IP with the distinguished set C. Hence R is empty and $|L| = r - 2 \ge m - 2$. This contradicts the acquired finite or assumed smaller-degree alternating bound.

Thus C is empty. If R were empty too, then $r = |L| + 1 \le m - 1$. Therefore R has one member and

$$r = m, \qquad |L| = m - 2. \tag{2}$$

The possibility $N = 0$ again contradicts the alternating bound for the faithful C-marking of A_m. The possibility $N = E$ gives $B = H$, since they have the same block image and $K \le E$, contrary to independence of the extra mark. The module list consequently forces

$$N = \langle z \rangle, \qquad K = E, \qquad m \text{ even}. \tag{3}$$

The last assertion follows because z must be even. The interval L, the one neighbour occupied by the omitted label, and the other neighbour occupied by R exhaust the whole string. After reversal,

$$i = 0, \qquad L = \{\rho_1, \dots, \rho_{m-2}\}, \qquad R = \{\rho_{m-1}\}.$$

The full parent diagram is connected: otherwise its commuting components and IP would give a nontrivial direct-product decomposition of the simple group A_{2m}. Every consecutive pair of marks in the interval therefore fails to commute.

The constant flip z centralizes H. The omitted ρ_0 does not commute with z, since otherwise z would centralize all of A_{2m}, whose natural centralizer is trivial. Put

$$T = \langle \rho_2, \dots, \rho_{m-2} \rangle.$$

This group centralizes ρ_0. If it contained z, the preceding noncommutation would be impossible. By (3) its block projection is therefore faithful. Denote its image by $Q \le A_m$. It has the inherited full C-marking of rank $m - 3$, with connected diagram. No quotient intersection property is assumed: the projection is an isomorphism on this actual subgroup.

If $m \ge 10$ and Q is transitive, the complete L047 theorem and $m - 3 > m/2 + 1$ force $Q = A_m$ or S_m. Evenness excludes S_m; the assumed or finite alternating rank bound excludes

A_m. If $m = 8$, the acquired complete maxima exclude A_8, and the sole proper transitive rank-five exception in L047 is the order-576 action of Lemma 2. This identification uses its actual accepted permutation marking: its five generators preserve

$$\{1, 2, 3, 8\} \mid \{4, 5, 6, 7\}$$

and are even. The even subgroup of that wreath stabilizer has order 576, the accepted exact order of the generated group, so they are equal. This uses the complete retained census and its actual coordinates, not an abstract group name alone. Lemma 2 makes ρ_0 preserve the original pairs, contradicting primitivity of the parent.

In every remaining case Q is intransitive. Apply Lemma 1 to its faithful lift T and the even involution ρ_0. It again preserves the original pairs. The group H also preserves them, so $\langle H, \rho_0 \rangle = A_{2m}$ would be imprimitive. This final contradiction proves the theorem. □

Attribution and boundary. This recovers the alternating block-image case of source Section 3.2. The source supplies the invariant-module reduction and the rank target; the new argument passes to the actual faithful endpoint subgroup and uses normal-factor fixed sets and point-stabilizer normalizers. It does not assert a splitting of the central extension B and never regards a block-breaking dihedral group as acting on the blocks. The complete acquired L047, symmetric extremal and small alternating classifications retain their mathematical and software premises. The full source, the symmetric block-image case, and the other general imprimitive cases remain separate obligations. This component adds no whole-original, open-question, Lean or compression credit.